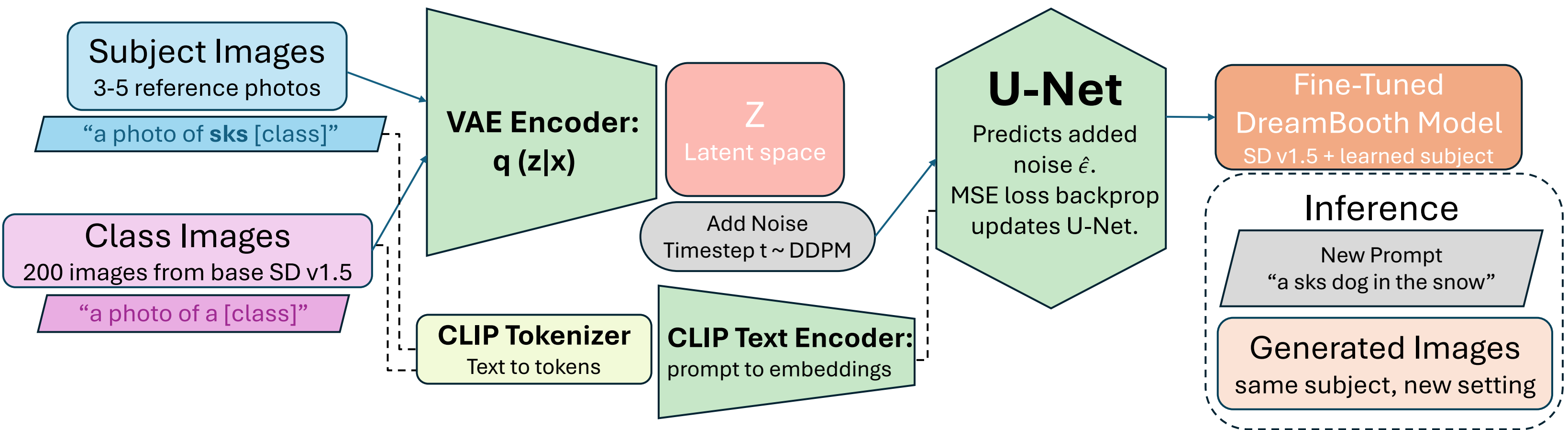


MOTIVATION

- Standard text-to-image models generate generic class members, not the same exact subject.
- DreamBooth teaches a pretrained model using only 3-5 reference images.
- Goal: preserve subject identity across new scenes, styles, and prompts.

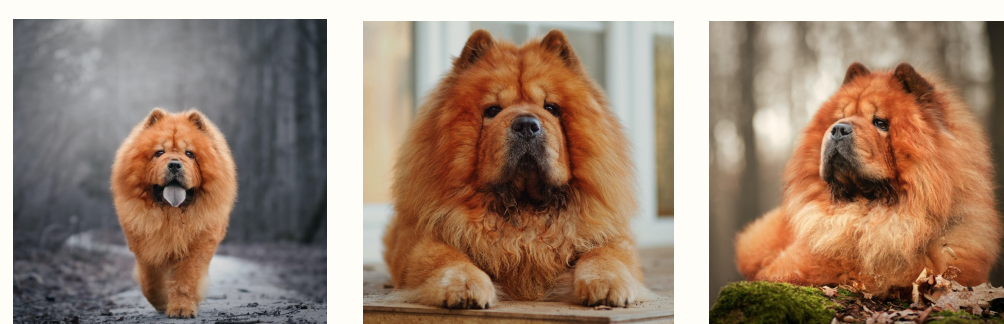
METHOD / TRAINING SETUP

Base model	Stable Diffusion v1.5
Components	VAE, CLIP text encoder, U-Net
Training	Frozen VAE and CLIP text encoder; fine-tuned only the U-Net
Instance images	3-5 reference photos with prompt: “a photo of sks [class]”
Class images	200 generated class images with prompt: “a photo of a [class]”
Image size	512 × 512
Loss	Noise-prediction MSE with prior preservation
Comparisons	Prior preservation on vs. off; learning rate 5e-6 vs. 3e-6, n-image sweep



CORE RESULT: SUBJECT-DRIVEN GENERATION

INSTANCE
“A photo of a **sks** dog...”

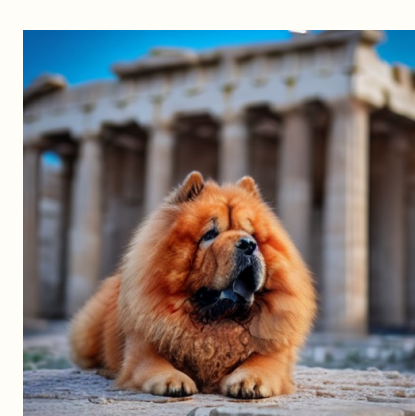


DreamBooth

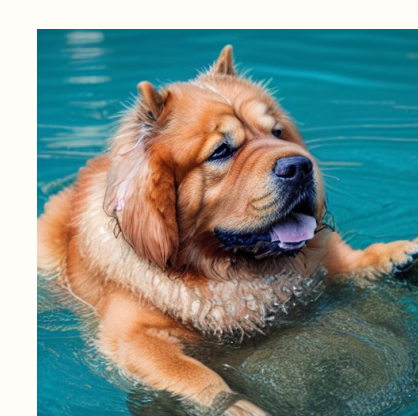
CLASS PRIOR
“A photo of a dog...”



“... in the Acropolis”



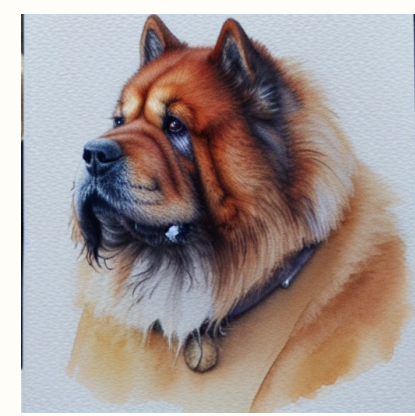
“... swimming underwater”



“a cross of a sks dog and a panda”



“...watercolor”



“... in a bucket”

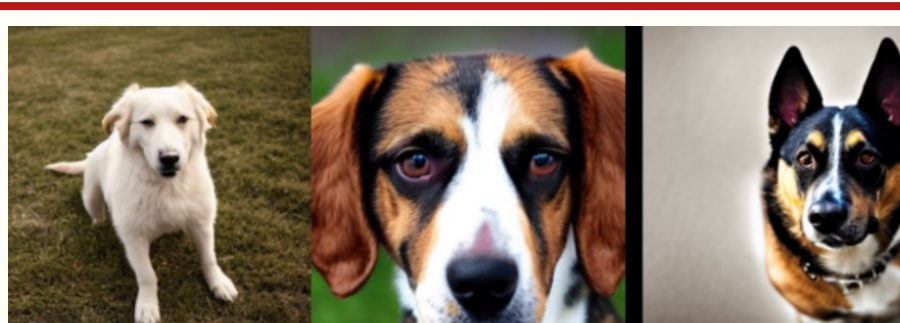


“a cross of a sks dog and a lion”

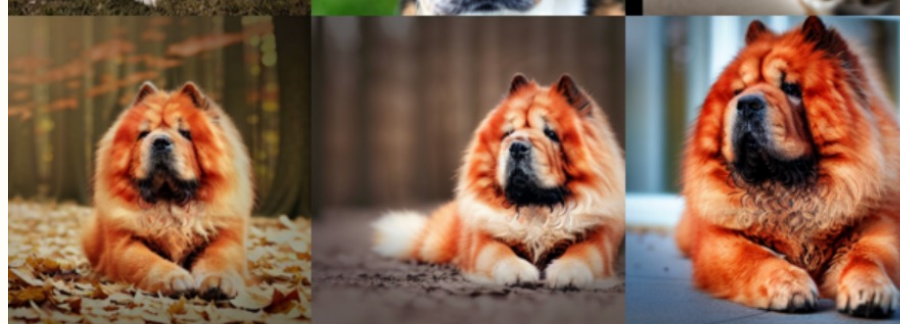


LANGUAGE DRIFT FROM REMOVING PRIOR PRESERVATION

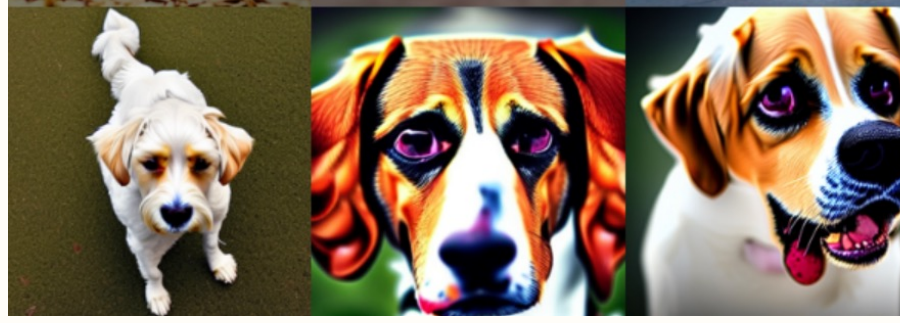
Before DreamBooth:
generic class prompt



Without prior loss



With prior loss



Teapot



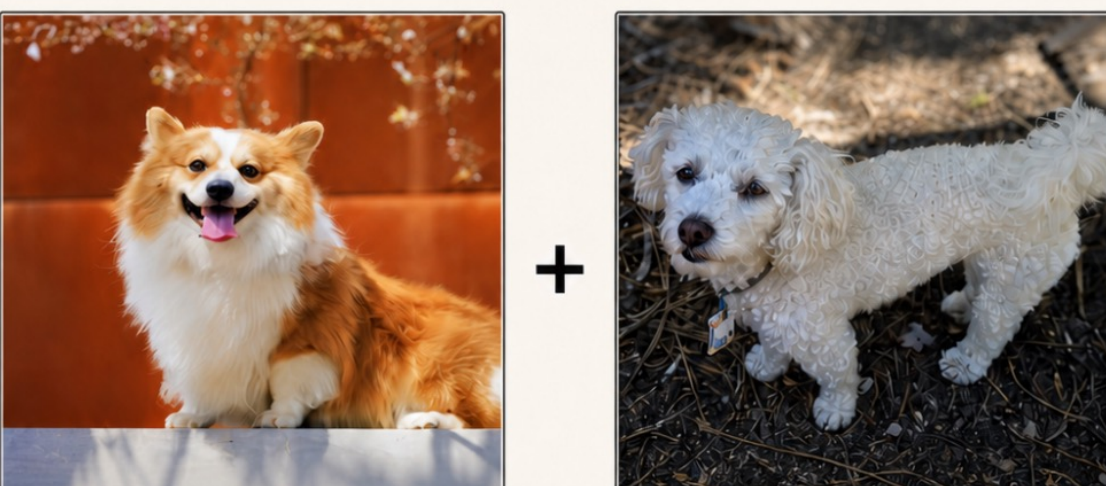
8. SUBJECT vs TEXT FIDELITY
DINO: “Does the new image look like my subject?”
CLIP-I: subject fidelity; general visual similarity
CLIP-T: “Does the image match the text prompt?”
PRES: “Does “a dog” still produce random dogs?”
DIV: “Are outputs varied or all the same?”

Result	DINO	CLIP-I	CLIP-T	Interpretation
Dreambooth Paper, Stable Diffusion	0.668	0.803	0.305	Paper baseline
Ours, LR = 5e-6	0.654	0.787	0.325	Best Overall Reproduction
Ours, LR = 3e-6	0.608	0.764	0.335	Higher prompt fidelity, lower subject fidelity

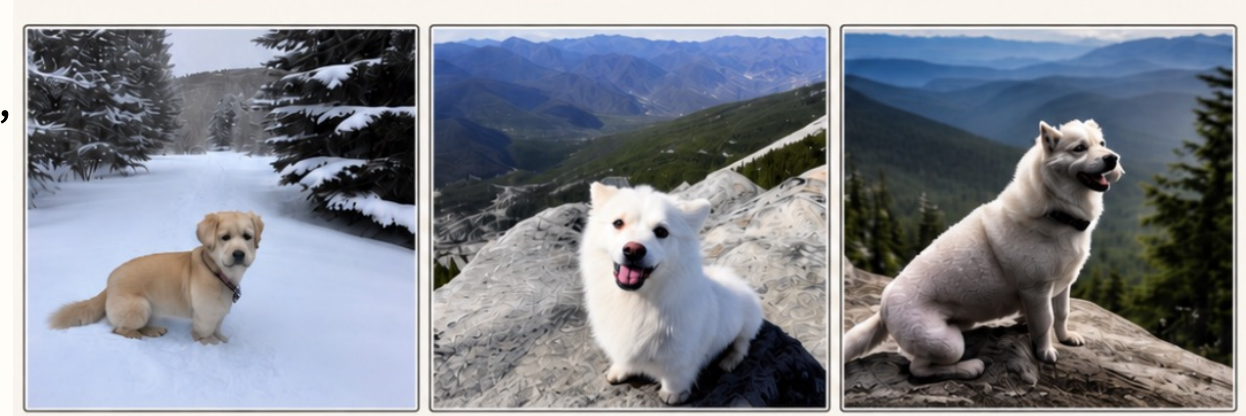
OTHER APPLICATIONS: DOG BREEDS

Shared-token training blends traits from **Dog 0** and **Dog 3** into new scenes.

Shared token: sks



“a photo of sks dog”



“...in the snow”

“...on a mountain”

LIMITATIONS

- Context leakage: training backgrounds sometimes appear in new prompts.
- Rare subjects were harder to model than common ones.
- Removing prior preservation caused class drift and overfitting.
- Our implementation used SD v1.5 on Colab, so results are not expected to exactly match the paper’s larger-model experiments.

TAKEAWAYS

- Prior preservation matters:** drift panel shows that without L_{prior} generic prompts like “a dog” collapse toward the learned subject.
- Identity and prompt fidelity trade off:** stronger subject identity risks less adherence to the prompt.

References:
1. “DreamBooth: Fine Tuning Text-To-Image Diffusion Models for Subject-Driven Generation.” *ArXiv:2208.12242 [Cs]*, 25 Aug. 2022. arxiv.org/abs/2208.12242.
2. “Google/Dreambooth.” *GitHub*, 20 Jan. 2024. github.com/google/dreambooth.

